

COMP 2121 – Quiz 4

Time: 20 min.

Possible marks: 5

Name:

Student #:

1. (3 marks) Use formula $\sum_{i=1}^n i = \frac{n(n+1)}{2}$ to find the value of counter after the following programming segment is executed

```
counter = 100;
for (i = 8; i <= n; i++){
    counter = counter + 2i
    for (j = i; j <= n+3; j++){
        counter = counter + 5
    }
}
//assume n>7
```

Simplify your answer and far as possible.

$$\begin{aligned} \text{counter} &= 100 + \sum_{i=8}^n \left(2i + \sum_{j=i}^{n+3} 5 \right) \\ &= 100 + \sum_{i=8}^n (2i + 5(n+3-i+1)) \\ &= 100 + \sum_{i=8}^n (-3i + 5n + 20) \\ &= 100 + \sum_{i=8}^n (-3i) + \sum_{i=8}^n (5n + 20) \\ &= 100 - 3 \left(\sum_{i=1}^n i - \sum_{i=1}^7 i \right) + (5n + 20)(n - 8 + 1) \\ &= 100 - 3 \left[\frac{n(n+1)}{2} - \frac{7(8)}{2} \right] + (5n + 20)(n - 7) \\ &= 100 - 3 \left(\frac{n^2 + n - 56}{2} \right) + 5n^2 + 20n - 35n - 140 \\ &= 100 - \frac{3n^2}{2} - \frac{3n}{2} + 84 + 5n^2 - 15n - 140 \\ &= \boxed{\frac{7n^2}{2} - \frac{33n}{2} + 44} \end{aligned}$$

2. (2 marks) Given

$$150 = 5b_{100} + \sum_{i=18}^{102} b_{i-2} + \sum_{j=7}^{14} b_{j+1} \quad \text{and} \quad \sum_{k=9}^{100} b_{k-1} = 108$$

find b_{100} .

$$150 = 5b_{100} + (b_{16} + b_{17} + \dots + b_{100}) + (b_8 + b_9 + \dots + b_{15}) \quad \text{and} \quad (b_8 + b_9 + \dots + b_{99}) = 108$$

$$150 = 6b_{100} + (b_8 + b_9 + \dots + b_{99}) \quad \text{and} \quad (b_8 + b_9 + \dots + b_{99}) = 108$$

$$150 = 6b_{100} + 108$$

$$42 = 6b_{100}$$

$$\boxed{b_{100} = 7}$$
